

Standard Form & Indices

Test 2 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 3^8 (1)

A2. 8^4 (1)

A3. 5^6 (1)

A4. $3^2 \times 3^3 = 3^5$ (1) = **243** (1)

Section B — Zero and negative indices

B1. 1 (1)

B2. $1 \div 4^2 = 1 \div 16$ (1) = **1/16** (1)

B3. $1 \div 10^4 = 1 \div 10000$ (1) = **0.0001** (1)

Section C — Writing standard form

C1. 5.4 (1) = **5.4×10^4** (1)

C2. 6.2 (1) = **6.2×10^6** (1)

C3. 6.7 (1) = **6.7×10^{-3}** (1)

C4. 9 (1) = **9×10^{-5}** (1)

Section D — Converting back, and calculating

D1. **4,200** (1)

D2. **0.0007** (1)

D3. $3 \times 3 = 9$, $10^2 \times 10^4 = 10^6$ (1) = 9×10^6 (1) already in standard form: **9×10^6** (1)

D4. $8 \div 2 = 4$, $10^7 \div 10^2 = 10^5$ (1) = 4×10^5 (1) already in standard form: **4×10^5** (1)

Section E — Powers to know, and estimating roots

E1. **64** (1)

E2. $9^2=81$, $10^2=100$ = between **9 and 10** (1)

Section F — Laws of indices with algebra

F1. p^9 (1)

F2. q^7 (1)

F3. r^{10} (1)

F4. $4 \times 3 = 12$ (1) $s^{3+4} = s^7$ (1) = **$12s^7$**

F5. $2^4 = 16$ (1) $t^{3 \times 4} = t^{12}$ (1) = **$16t^{12}$**

Section G — Fractional indices (Higher)

G1. $\sqrt{36} = \mathbf{6}$ (1)

G2. cube root of 8 = $\mathbf{2}$ (1)

G3. 4th root of 81 = 3 (1) $3^3 = \mathbf{27}$ (1)

G4. $\sqrt{16} = 4$ (1) $1 \div 4 = \mathbf{1/4}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.